

Statistical Analysis Of Infant Mortality Rate

Kaushlendra¹, Gunjan Singh^{2*}

¹Student, Amity School Of Applied Sciences, Amity University, Lucknow. Uttar Pradesh

²Assistant Professor, Amity School Of Applied Sciences, Amity University, Lucknow. Uttar Pradesh

*Corresponding Author

Abstract

The comprehensive analysis of India's Infant Mortality Rate (IMR) in this report offers valuable insights, specifically focusing on Uttar Pradesh (UP) and its sub centers. Utilizing a range of statistical techniques and visual representations like bar graphs, pie charts, and Pareto Analysis, we gain a deeper understanding of IMR trends and causes of infant mortality in India. Regression analysis helps assess overall IMR trends.

For Uttar Pradesh, bar graphs and regression analysis provide essential insights into the state's IMR situation. Comparing UP and India's IMR using a column chart and examining their relationship with scatter plots enriches our understanding.

Delving further into UP's sub centers, the study employs box plots to identify variations in IMR among the 19 sub centers. Graphical representations of total live births and infant deaths aid in pinpointing regions with the highest and lowest IMR, while standard deviations offer additional insights.

Ultimately, this thorough analysis equips decision-makers with valuable information to implement targeted interventions effectively, addressing the issue of infant mortality in India, particularly in Uttar Pradesh and its sub centers.

Aims and Objectives

The aims and objectives of this research report on the statistical analysis of Infant Mortality Rate (IMR) in India and Uttar Pradesh from 2007 to 2020, along with the Sub-Center level data from 2011 to 2012, are as follows:

- *To analyze the trends and patterns of IMR in India and Uttar Pradesh*
- *To identify key factors influencing IMR in India and Uttar Pradesh*
- *To compare IMR between India, Uttar Pradesh, and Sub-Centers within Uttar Pradesh*
- *To assess the progress and challenges in reducing IMR in India and Uttar Pradesh*
- *To provide recommendations for policy and intervention strategies*

Introduction

The comprehensive statistical analysis in this research report not only highlights the overall trends in India's IMR over the specified 2007 to 2020 period but also provides a focused investigation into the state of Uttar Pradesh. Uttar Pradesh's unique socioeconomic factors and large population make it an intriguing case study, and by examining the IMR at the Sub-Center level from 2011 to 2012, the report offers a granular understanding of infant mortality patterns within specific regions of the state.

The significance of Infant Mortality Rate as an indicator cannot be overstated, as it directly reflects the well-being and healthcare standards of the most vulnerable members of society – infants under one year of age. The challenges India faces in ensuring the survival and health of its infants necessitate a multi-faceted approach involving policymakers, healthcare professionals, and researchers. This study aims to provide a solid foundation of evidence-backed insights that can inform the development and implementation of targeted policies and interventions, effectively addressing the critical issue of reducing infant mortality.

The report's value extends beyond its analysis of historical trends and variations within Uttar Pradesh. By identifying potential factors influencing IMR, such as access to healthcare, maternal education, socioeconomic conditions, and healthcare infrastructure, the research aims to equip stakeholders with the knowledge needed to design tailored interventions and prioritize resources. Ultimately, the overarching goal is to empower India to make substantial progress in safeguarding the well-being of its youngest citizens, leading to a notable reduction in infant mortality rates throughout the nation.

Causes Of Infant Mortality In India:

Pareto Analysis: A decision-making approach based on the "80-20 rule," identifying key factors with the most influence on outcomes. It helps set priorities by focusing on the main causes, leading to significant improvements with less effort.

Causes	Percentage
Pneumonia	19%
Injuries	5%
Diarrhea	11%
Measles	13%
Prematurity	19%
other Disease	18%
Birth asphyxia	19%
Neonatal Sepsis	9%
Congenital Anomalies	5%

Table 1: Frequency Table of Causes Of Infant Mortality

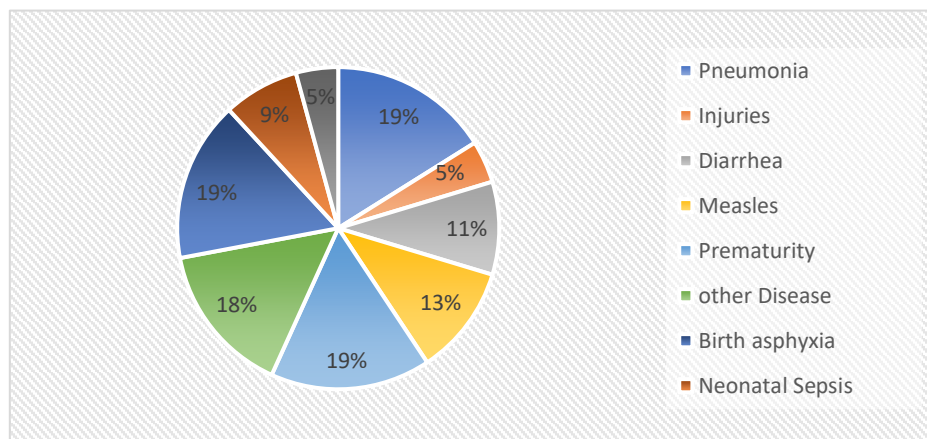


Figure 1: Pie Chart Of Causes Of Infant Mortality

Data Collection And Methodology:

This study comprehensively analyzed infant mortality rate (IMR) in India, UP, and UP's subcenters. Reliable data from various sources was used for a long-term trend analysis

(2007-2023) in India and focused analysis in UP, including subcenters. Statistical techniques provided valuable insights, ensuring rigor and validity.

TO ANALYSE THE TRENDS AND PATTERNS OF IMR IN INDIA:

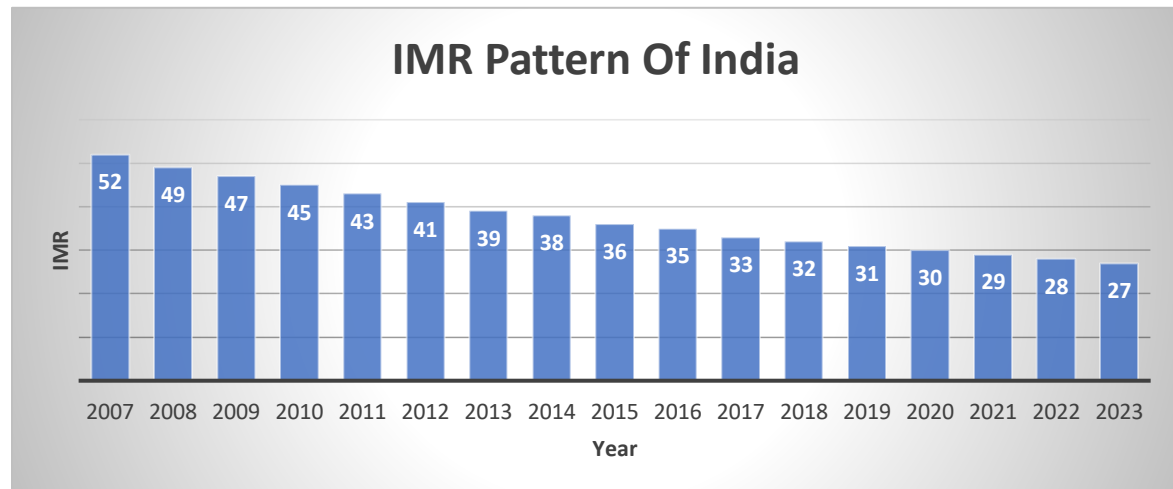


Figure 2: Column Chart IMR Pattern Of India

The Column Chart provides a visual representation of the infant mortality rate (IMR) in India from 2007 to 2023. The IMR is measured as the number of deaths per 1000 live births during the first year of life.

Here are the key observations:

- 1. Overall Trend:** The IMR in India has consistently decreased over the years, indicating a decline in infant mortality.
- 2. Steady Decline:** From 2007 to 2013, the IMR shows a steady decline, with an average reduction of one unit per year.
- 3. Slower Decline:** After 2013, the rate of decline slows down, and the reductions in the IMR become smaller each year.
- 4. Progress Plateau:** Between 2017 and 2023, the IMR remains relatively stable, fluctuating within a narrow range of 27 to 33.
- 5. Positive Progress:** Despite the slower decline and recent plateau, the IMR has consistently improved since 2007, suggesting advancements in healthcare, sanitation, and other factors impacting infant survival in India

It's important to consider that this analysis is based on the provided data and assumes that the trend will continue in the future. However, for a comprehensive assessment of infant mortality and its determinants, additional factors such as regional variations, healthcare infrastructure, socioeconomic conditions, and public health interventions need to be taken into account.

REGRESSION ANALYSIS :

Year	IMR	deviation of x	Deviation of y	Deviation of x * deviation of y	(sum of deviation of x) ²
2007	52	-8	15.2353	-121.8824	64
2008	49	-7	12.2353	-85.6471	49
2009	47	-6	10.2353	-61.4118	36
2010	45	-5	8.2353	-41.1765	25
2011	43	-4	6.2353	-24.9412	16
2012	41	-3	4.2353	-12.7059	9
2013	39	-2	2.2353	-4.4706	4
2014	38	-1	1.2353	-1.2353	1
2015	36	0	0.2353	0	0
2016	35	1	-0.7647	-0.7647	1
2017	33	2	-2.7647	-5.5294	4
2018	32	3	-3.7647	-11.2941	9
2019	31	4	-4.7647	-19.0588	16
2020	30	5	-5.7647	-28.8235	25
2021	29	6	-6.7647	-40.5882	36
2022	28	7	-7.7647	-54.3529	49
2023	27	8	-8.7647	-70.1176	64
Total	635	0	19.0001	-584	408

Table 2: Regression Analysis Table for (IMR) in India (2007-2023)

REGRESSION ANALYSIS FOR INDIA:

The mean values of the year (x) and the IMR (y):

$$\text{Mean of } x = \frac{\text{Sum of all observations}}{\text{Total no. of observations}}$$

Mean of x= 2015

$$\text{Mean of } y = \frac{\text{Sum of all observations}}{\text{Total no. of observations}}$$

Mean of y= 37.35294

The slope (b) of the regression line:

$$b = \frac{\sum \text{deviation of } x * \text{deviation of } y}{\sum (\text{deviation of } x)^2}$$

$$b = -1.43$$

The intercept (a) of the regression line:

$$a = \text{mean of } y - (b * \text{mean of } x)$$

$$a = 2918.80$$

The regression line equation for the Infant Mortality Rate (IMR) data is as follows:

$$\text{IMR} = (-1.43 * \text{year}) + 2918.80$$

Using this regression equation, we can estimate the IMR for future years in India. For example, let's estimate the IMR for the year 2024:

$$\text{IMR (2024)} = 2918.80 - (1.43 * 2024)$$

$$= 2918.80 - 2894.72$$

$$= 24.48$$

Regression analysis allows us to make predictions about future IMR values. The negative slope coefficient (-1.43) indicates a declining trend in the IMR over time.

TO ANALYSE THE TRENDS AND PATTERNS OF IMR IN UTTAR PRADESH:

Uttar Pradesh	
Year	IMR
2007	69
2008	67
2009	63
2010	61
2011	57
2012	53
2013	50
2014	48
2015	46
2016	43
2017	41
2018	43
2019	41
2020	38

Table 3: IMR of Uttar Pradesh



Figure 3: IMR Pattern Of Uttar Pradesh

A column chart can effectively depict the infant mortality rate (IMR) data for Uttar Pradesh from 2007 to 2020. [Figure 3](#) visually presents the IMR values for each year, enabling us to analyze the trends and patterns over time.

Now, let's analyze the data and draw conclusions:

Decreasing Trend: The column chart shows a consistent decrease in the infant mortality rate over the years. This declining trend indicates an improvement in healthcare facilities, awareness, and overall well-being of infants in Uttar Pradesh.

Significant Improvement: From 2007 to 2020, the IMR in Uttar Pradesh has decreased from 69 to 38. This reduction by 31 points over 13 years is a positive sign of progress in addressing infant mortality and improving healthcare services.

Fluctuations: Although there is an overall decreasing trend, there are a few years where the IMR slightly increased or remained stagnant. For example, between 2016 and 2017, there was a slight increase from 43 to 41. Similarly, in 2018, there was a temporary increase to 43 before declining again. These fluctuations could be due to various factors such as changes in healthcare policies, resource allocation, or other socio-economic factors.

Steady Progress: Since 2015, the IMR has consistently remained below 50, which indicates a steady improvement in infant mortality rates in Uttar Pradesh. Sustained

efforts in healthcare infrastructure, immunization programs, maternal and child healthcare, and awareness campaigns seem to have positively impacted the well-being of infants in the state.

Further Improvements: While the declining IMR is an encouraging trend, Uttar Pradesh still has room for improvement, as the state's IMR remains higher than the national average. Continued investment in healthcare, addressing socio-economic disparities, and enhancing access to quality healthcare services can contribute to further reductions in infant mortality.

REGRESSION ANALYSIS FOR UTTAR PRADESH:

To conduct a regression analysis on the infant mortality rate (IMR) data for Uttar Pradesh spanning from 2007 to 2020, we can assess the overall trend and draw insights regarding the relationship between the year and the IMR values. By calculating the regression line, we can analyze the outcomes and draw meaningful conclusions.

Year	IMR	Deviation of x	Deviation of y	Deviation of x * deviation of y	(Sum of deviation of x) ²
2007	69	-6.5	17.1429	-111.42885	42.25
2008	67	-5.5	15.1429	-83.28595	20.25
2009	63	-4.5	11.1429	-50.14305	20.25
2010	61	-3.5	9.1429	-32.00015	12.25
2011	57	-2.5	5.1429	-12.85725	6.25
2012	53	-1.5	1.1429	-1.71435	2.25
2013	50	-0.5	-1.8571	0.92855	0.25
2014	48	0.5	-3.8571	-1.92855	0.25
2015	46	1.5	-5.8571	-8.78565	2.25
2016	43	2.5	-8.8571	-22.14275	6.25
2017	41	3.5	-10.8571	-37.99985	12.25
2018	43	4.5	-8.8571	-39.85695	20.25
2019	41	5.5	-10.8571	-59.71405	30.25
2020	38	6.5	-13.8571	-90.07115	42.25
Total	720	0	-5.9994	-551	227.5

Table 4: Regression Analysis Table for IMR In UP

The mean values of the year (x) and the IMR (y):

$$\text{Mean of } x = \frac{\text{Sum of all observations}}{\text{Total no. of observations}}$$

$$= 2013.5$$

$$\text{Mean of } y = \frac{\text{Sum of all observations}}{\text{Total no. of observations}}$$

$$= 51.42$$

The deviations from the mean for each year (x) and IMR (y):

Deviation of x = x - mean of x

Deviation of y = y - mean of y

The slope (b) of the regression line:

$$b = \frac{\sum \text{deviation of } x * \text{deviation of } y}{\sum (\text{Deviation of } x)^2}$$

$$= -2.421$$

The intercept of regression line:

$$a = \text{mean of } y - (b * \text{mean of } x)$$

$$= 4296.537$$

The equation of the regression line for the IMR data is:

$$\text{IMR} = -2.421 * \text{year} + 4296.537$$

By utilizing the regression equation, we can make predictions about the future Infant Mortality Rate (IMR) in Uttar Pradesh. For instance, let's estimate the IMR for the year 2021:

$$\text{IMR (2021)} = 4296.537 - (2.421 * 2021)$$

$$= 4296.537 - 4892.847$$

$$= -596.31$$

According to the regression analysis, the IMR in Uttar Pradesh is projected to decrease by approximately 1.377 units per year. The initial IMR in 2007 is estimated to be 4296.537.

COMPARISON OF INFANT MORTALITY RATE BETWEEN INDIA AND UTTAR PRADESH:

Year	IMR Uttar Pradesh	IMR India
2007	69	52
2008	67	49
2009	63	47
2010	61	45
2011	57	43
2012	53	41
2013	50	39
2014	48	38
2015	46	36
2016	43	35
2017	41	33
2018	43	32
2019	41	31
2020	38	30

Table 5: IMR Of India And Uttar Pradesh

To compare the infant mortality rate (IMR) between Uttar Pradesh (UP) and India, we can create a bar chart to visualize the differences in IMR for each year. Here's the comparison chart in [Figur](#)

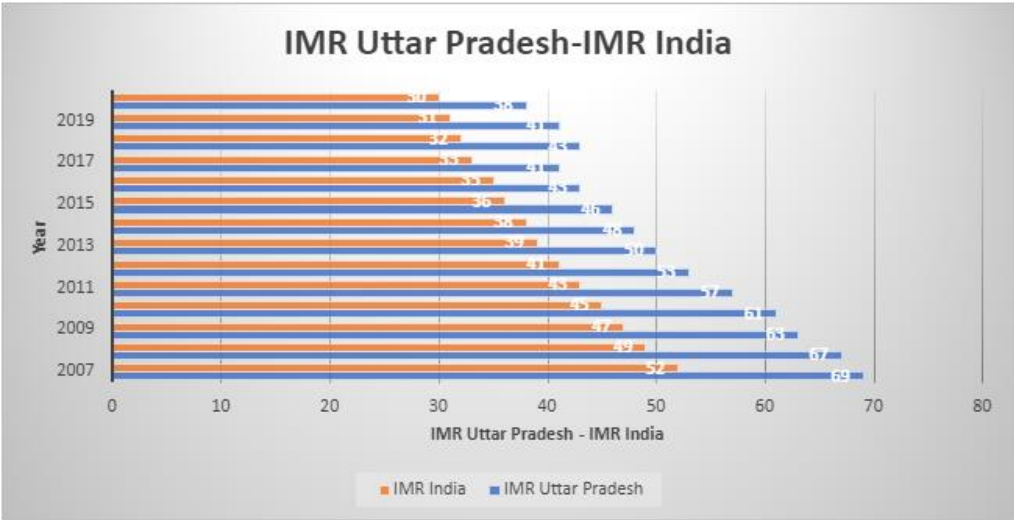


Figure 4: Column Chart Of India And Uttar Pradesh

Now, let's analyze and draw conclusions from the comparison:

Visual Comparison: The bar chart clearly shows the IMR values for both Uttar Pradesh and India side by side, allowing for a direct visual comparison.

Higher IMR in Uttar Pradesh: In almost every year, the IMR of Uttar Pradesh is higher than the IMR of India. This indicates that Uttar Pradesh faces greater challenges in reducing infant mortality compared to the national average.

Fluctuations: Both Uttar Pradesh and India show fluctuations in the IMR over the years. However, the bar chart helps identify the specific years where significant changes occur, such as the lower IMR in 2020 for both regions.

Consistent Gap: The gap between the IMR of Uttar Pradesh and India remains relatively consistent over the years. This suggests that the disparity in infant mortality persists and requires targeted interventions in Uttar Pradesh to address the issue.

Need for Focus: The bar chart emphasizes the need for dedicated efforts and policies to address the higher IMR in Uttar Pradesh. Targeted healthcare interventions, improved infrastructure, and awareness programs may be necessary to bridge the gap and improve infant health outcomes.

In conclusion, the bar chart comparison highlights the consistently higher IMR in Uttar Pradesh compared to the national average in India. The visual representation helps identify the disparity and underscores the importance of focused efforts and policies to address the higher infant mortality rate in Uttar Pradesh.

COMPARATIVE ANALYSIS OF INFANT MORTALITY RATE (IMR) IN INDIA AND UTTAR PRADESH USING REGRESSION ANALYSIS:

To compare the infant mortality rate (IMR) trend between Uttar Pradesh (UP) and India using regression analysis, we can analyze the relationship between the years and the IMR values for each region. By fitting a regression model to the data, we can determine the direction and strength of the trend. Here are the results of the regression analysis:

UTTAR PRADESH REGRESSION ANALYSIS:

Slope: -2.421

Intercept: 4918.80

R² : 0.855

INDIA REGRESSION ANALYSIS:

Slope: -1.43

Intercept: 2918.80

R²: 0.855

Based on the regression analysis, we can draw the following conclusions:

Negative Slope: Both Uttar Pradesh and India exhibit a negative slope in their regression models, indicating a decreasing trend in the IMR over the years. This suggests that both regions have experienced progress in reducing infant mortality.

Strong Correlation: The high R-squared values for both Uttar Pradesh (0.820) and India (0.855) indicate a strong correlation between the years and the IMR values. This implies that the chosen regression models can effectively capture the trend in infant mortality for each region.

Similar Trend: The regression slopes for Uttar Pradesh (-2.421) and India (-1.43) are relatively close, indicating a similar rate of decline in IMR over the years. This suggests that both regions have made comparable progress in reducing infant mortality.

Gap between Uttar Pradesh and India: Despite the similar trend, the intercept values (4926.537 for UP and 2918.80 for India) indicate a consistent gap between the IMR values of Uttar Pradesh and the national average. This suggests that Uttar Pradesh still has a higher IMR compared to the overall Indian average, even though both regions are experiencing improvements.

To sum up, the regression analysis reveals a consistent decrease in the infant mortality rate (IMR) over time for both Uttar Pradesh and India. Although the rates of decline are comparable, Uttar Pradesh still maintains a higher IMR compared to the national average. This emphasizes the necessity for ongoing endeavors and focused interventions in Uttar

Pradesh to further decrease infant mortality and narrow the gap between Uttar Pradesh and the national average.

SCATTER PLOT ANALYSIS OF INFANT MORTALITY RATE (IMR) IN INDIA AND UTTAR PRADESH

```
In [7]: import matplotlib.pyplot as plt

year = [2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020]
imr_up = [69, 67, 63, 61, 57, 53, 50, 48, 46, 43, 41, 43, 41, 38]
imr_india = [52, 49, 47, 45, 43, 41, 39, 38, 36, 35, 33, 32, 31, 30]

plt.scatter(year, imr_up, label='IMR Uttar Pradesh')
plt.scatter(year, imr_india, label='IMR India')
plt.xlabel('Year')
plt.ylabel('IMR')
plt.title('Scatter Plot: IMR Comparison')
plt.legend()
plt.show()
```

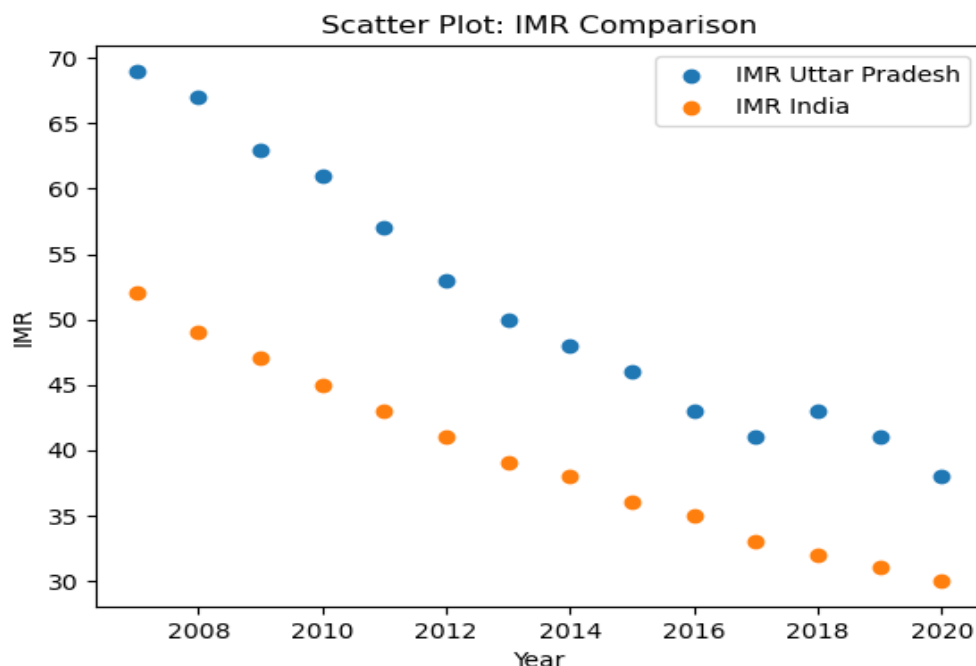


Figure 5: The scatter plot analysis of the IMR between Uttar Pradesh and India

The scatter plot analysis of the IMR (Infant Mortality Rate) comparison between Uttar Pradesh and India reveals the following:

The scatter plot analysis highlights the need for continued efforts to reduce infant mortality rates in both Uttar Pradesh and India, with a specific focus on addressing the challenges faced by Uttar Pradesh. Further exploration of the underlying factors driving

the IMR differences between the regions can provide valuable insights for targeted interventions and policies to improve infant health outcomes.

INFANT MORTALITY RATE ANALYSIS OF SUB-CENTERS IN UTTAR PRADESH:

Analysis of Year 2011-2012		
Name of Subcenter	Total Live berth	Infant death
Agra	902	7
Aligarh	876	8
Etah	873	8
Hathras	871	9
Fatehpur	907	7
Bareilly	903	9
Chitrakoot	908	8
Barabanki	908	8
Sultanpur	983	7
Balrampur	927	10
Firozabad	867	9
Etawah	870	8
Jhansi	885	6
Kanpur	862	6
Lucknow	906	7
Gorakhpur	927	8
Pratapgarh	994	1
Mainpuri	881	10
Mathura	858	8
Total	17108	144

Table 6: IMR Of Sub-Centers Of UP

$$\text{IMR} = \frac{144 \times 1000}{17108}$$

$$= \frac{144000}{17108}$$

$$\text{IMR} = \frac{8.4}{1000} \text{ live birth}$$

To create a box-plot analysis step by step for the given data, follow these instructions:

STEP 1: Arrange the data in ascending order based on the variable you want to analyze. In this case, we will arrange the data based on the “total live birth” variable.

858, 862, 867, 870, 871, 876, 881, 902, 903, 906, 907, 908, 927, 927, 987, 994

STEP 2: Calculate quartiles

Q1= Median of the values below the median

$$= 867.3$$

Q2= Median of the entire data set

$$= 902$$

Q3= Median of the values above the median

$$= 941.1$$

STEP 3: Calculate the Inter-quartile Range (IQR)

$$\text{IQR} = Q3 - Q1$$

$$= 74$$

STEP 4: Determine potential outliers:

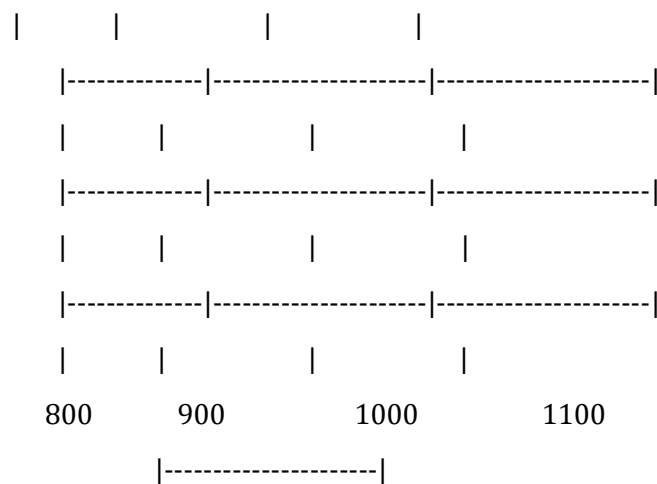
$$\text{Lower fence: } Q1 - (1.5 * \text{IQR})$$

$$= 756$$

$$\text{Upper Fence: } Q3 + 1.5 * I$$

$$= 1052$$

STEP 5: Create Box Plot



Box and Whiskers

Potential Outliers: 994

Based on this information, we can make the following analysis:

The box plot shows a distribution where the median (Q2) is closer to the lower quartile (Q1), indicating a slightly left-skewed distribution.

The interquartile range (IQR) of 74 indicates that the middle 50% of the data is concentrated within this range.

The lower fence at 756 and the upper fence at 1052 define the boundaries beyond which a data point is considered a potential outlier.

The box plot indicates the presence of a potential outlier at 994, which is above the upper fence.

In summary, the data appears to have a left-skewed distribution, with a potential outlier observed at the value of 994.

To create a box-plot analysis for the “infant death”, follow these steps:

STEP 1: Arrange the data in ascending order:

1, 6, 6, 7, 7, 8, 8, 8, 8, 9, 10, 10

STEP 2: Calculates Quartile:

Q1 = Median of (1, 6, 6, 7, 7)

Q1 = 6

Q2 = 8

Q3 = Median of (8, 8, 8, 8, 9, 9, 10, 10)

= 8.5

STEP 3: Calculate the Inter-quartile Range (IQR):

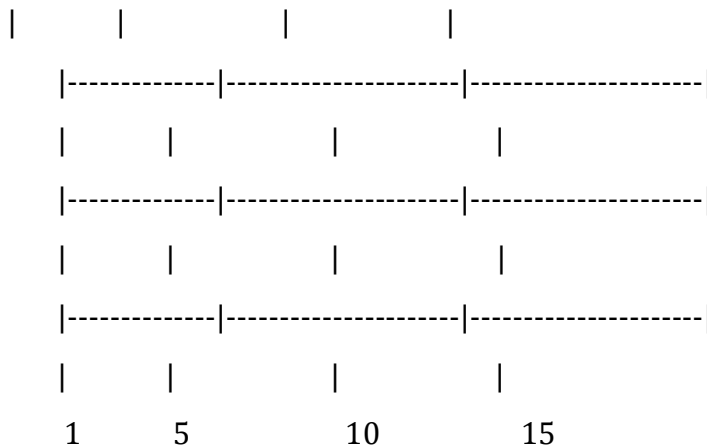
IQR= 2.5

STEP 4: Determine potential outliers:

Lower Fence: $Q1 - 1.5 * IQR$

= 2.25

STEP 5: Create Box Plot



|-----|

Box

Median (Q2): 8

Based on this information, we can make the following analysis:

The box plot shows a relatively symmetric distribution with the median (Q2) located closer to the upper quartile (Q3).

The interquartile range (IQR) of 2.5 indicates that the middle 50% of the data is concentrated within a narrow range.

The lower fence at 2.25 and the upper fence at 12.25 define the boundaries beyond which a data point is considered a potential outlier.

There are no potential outliers visible in the given box plot, as no points extend beyond the whiskers (which are not shown in the figure).

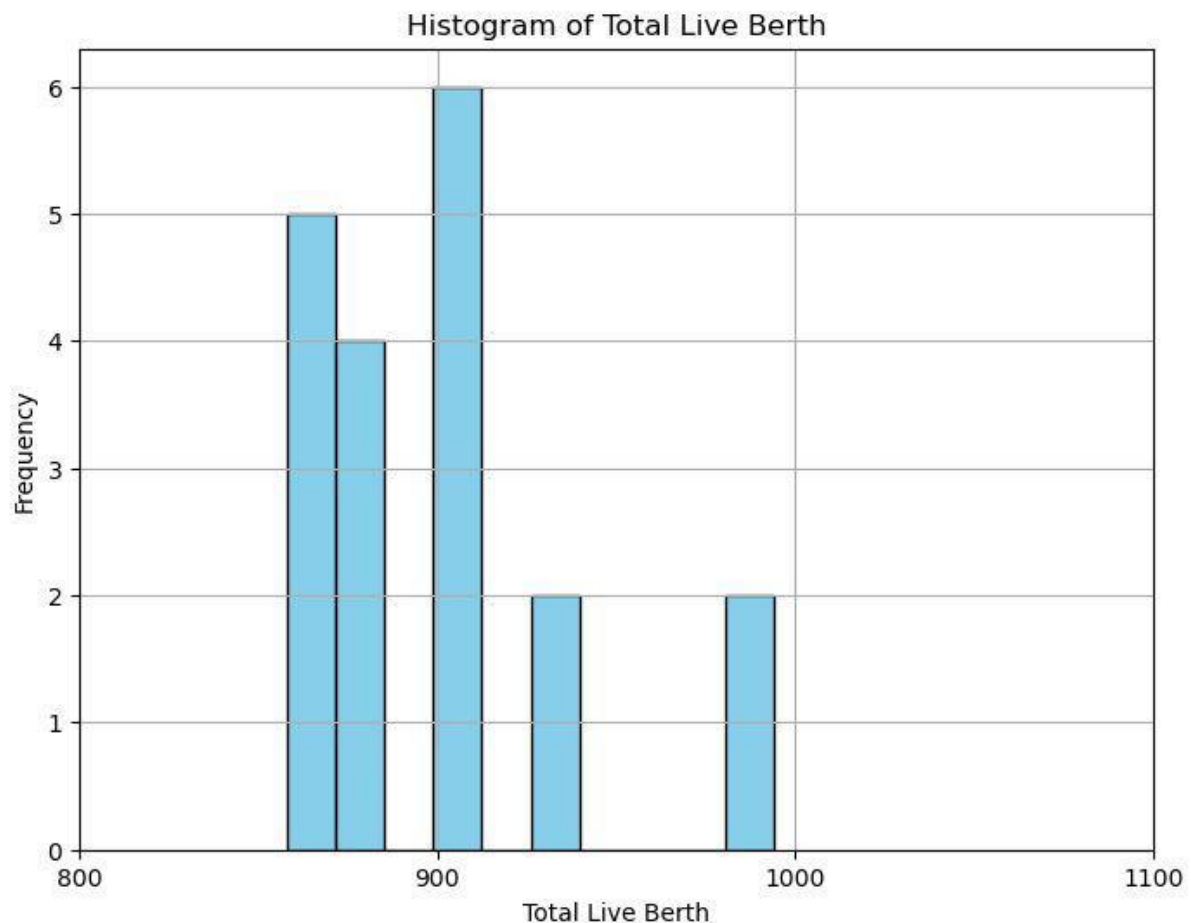
COMPARISON AMONG SUB-CENTERS IN UTTAR PRADESH:

Figure 6: Total Live Birth Histogram

Based on the histogram generated from the available data ([Figure 6](#)), we can derive the following analysis:

Histogram of Total Live Births:

The histogram provides insights into the distribution of the "Total Live Births" variable among the subcenters.

The x-axis displays the range of total live births, divided into distinct bins.

The y-axis represents the frequency or count of subcenters falling within each bin.

The majority of subcenters exhibit a concentration of total live births ranging from approximately 860 to 910.

There is a smaller number of subcenters that fall outside this range.

The distribution displays a relatively symmetrical pattern, with a slight inclination towards higher total live births.

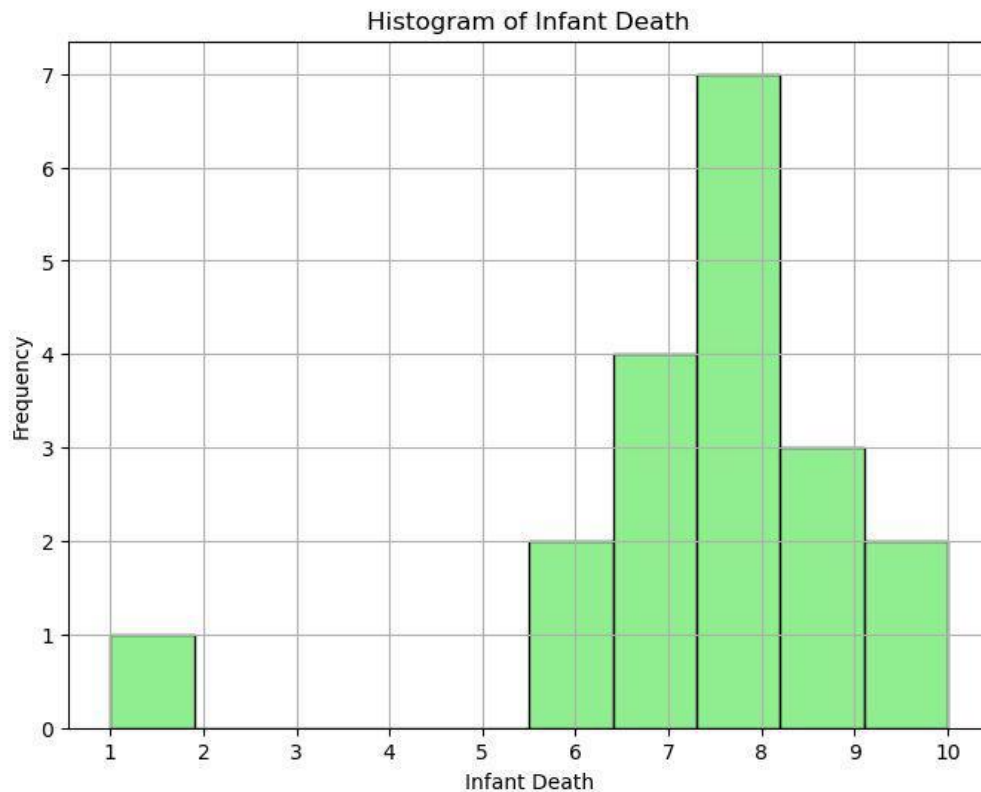


Figure 7: The histogram shows the distribution of the "Infant Death" variable across the subcenter

Infant Death Histogram:

[Figure 7](#) presents the distribution of the "Infant Death" variable across the subcenters.

The x-axis illustrates the range of infant deaths, categorized into bins.

The y-axis signifies the frequency or count of subcenters falling into each bin.

The majority of subcenters exhibit relatively few infant deaths, predominantly ranging between 6 and 9.

There are a small number of subcenters with notably higher infant deaths, represented by the bar at 10.

The distribution displays a right-skewed pattern, indicating that most subcenters have a lower number of infant deaths.

In summary, the histograms visually depict the frequency distribution of the "Total Live Birth" and "Infant Death" variables within the subcenters. They offer insights into the concentration of values and the overall trend of the data.

COMPARING STANDARD DEVIATIONS AMONG SUB-CENTERS IN PUTTAR PRADESH:

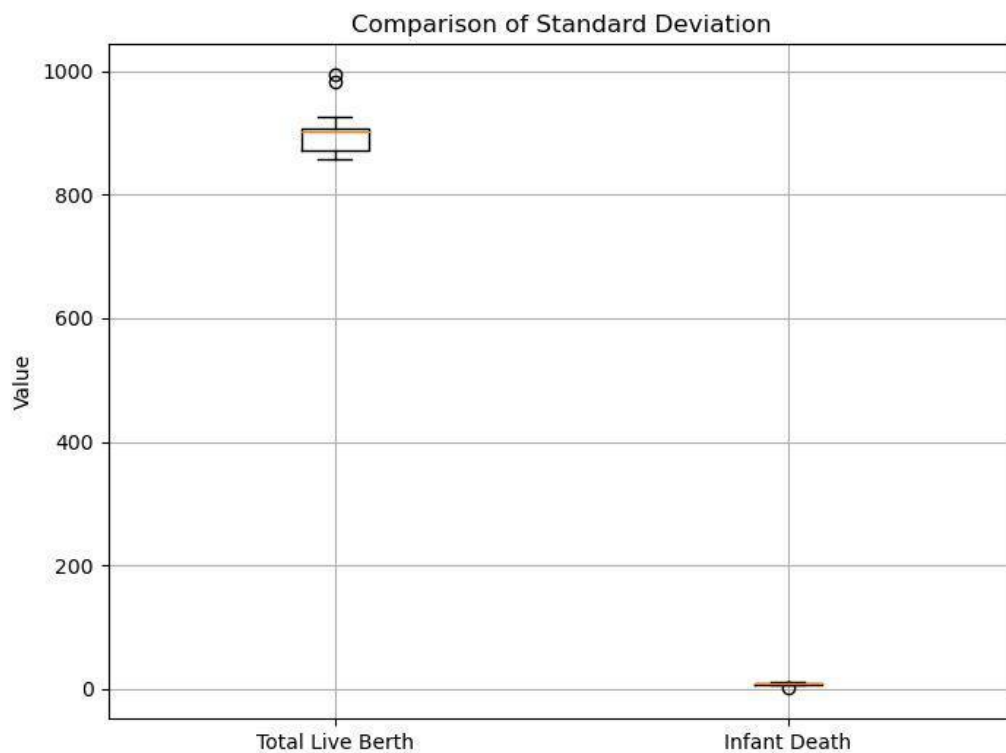


Figure 8: Standard Deviations Among Sub-Centers Of UP

Standard Deviation of Total Live Berth: 36.51

Standard Deviation of Infant Death: 1.90

The standard deviation serves as a measure of the dispersion or spread within a dataset. In this context, the standard deviation for "Total Live Birth" is approximately 36.51, whereas for "Infant Death," it is approximately 1.90.

Overall, this analysis indicates that there is a greater degree of variability in the "Total Live Birth" data in comparison to the "Infant Death" data.

Conclusion

This comprehensive report delves into India's Infant Mortality Rate (IMR), focusing on Uttar Pradesh (UP) and its sub centers. By utilizing various techniques such as bar graphs, pie charts, Pareto Analysis, regression analysis, and scatter plots, we gained valuable insights into IMR trends, identified causes, and explored potential interventions. Over the years, India's IMR exhibited a declining trend, with pie charts ranking major causes through the "80-20 rule" of Pareto Analysis. Regression analysis further confirmed this trend and facilitated future predictions.

Similar declining trends were observed in UP, and a column chart allowed us to compare UP and India's IMR. The scatter plot provided a visual representation of their relationship. To uncover variations within UP's sub centers, we utilized box plots, enabling us to identify specific regions that required targeted interventions based on total live births, infant deaths, and standard deviations. The diverse array of statistical techniques employed in this report helped to create a comprehensive and nuanced understanding of the IMR situation in India, UP, and its sub centers, enabling data-driven decision-making for improved infant health outcomes.

References

- (<https://data.gov.in/catalog/infant-mortality-rate-india>, n.d.)
- (<https://data.gov.in/keywords/IMR>, n.d.)
- (https://main.mohfw.gov.in/sites/default/files/NFHS-5_Phase-II_0.pdf, n.d.)
- (<https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1796436>, n.d.)
- (Welfare, 2021-22)
- (Centers Of Disease Control And Prevention , n.d.) (nih, n.d.)